**JHU Engineering for Professionals**
**Applied and Computational Mathematics**
**Data Mining 625.740**

**Homework for Module on Linear Discrimination**

## Theoretical Problems (1 point each)

1. (a) Show that the distance from the hyperplane $g(\mathbf{x}) = \mathbf{w^T x} + w_0 = 0$ to the point $\mathbf{x}_a$ is $|g(\mathbf{x}_a)|/||\mathbf{w}||$ by minimizing $||\mathbf{x} - \mathbf{x}_a||^2$ subject to the constraint $g(\mathbf{x}) = 0$.

   (b) Show that the projection of $\mathbf{x}_a$ onto the hyperplane is given by

$$\mathbf{x_p} = \mathbf{x}_a - \frac{g(\mathbf{x}_a)}{||\mathbf{w}||^2}\mathbf{w}.$$

2. Let $\mathbf{x_1}, \ldots, \mathbf{x_n}$ be $n$ $q$-dimensional samples and $Q$ be any nonsingular positive definite $q \times q$ matrix. Show that the vector $\mathbf{x}$ that minimizes

$$\sum_{k=1}^{n} (\mathbf{x_k} - \mathbf{x})^T Q^{-1} (\mathbf{x_k} - \mathbf{x})$$

   is the sample mean, $\bar{\mathbf{x}} = \frac{1}{n}\sum_{k=1}^{n} \mathbf{x_k}$.

3. Consider a linear classifier with discriminant functions $g_i(\mathbf{x}) = \mathbf{w_i^T x} + w_{i0}, \quad i = 1, \ldots, c$. Show that the decision regions are convex by showing that if $\mathbf{x}_1 \in \mathcal{R}_i$ and $\mathbf{x}_2 \in \mathcal{R}_i$ then $\lambda \mathbf{x}_1 + (1 - \lambda)\mathbf{x}_2 \in \mathcal{R}_i$ if $0 \leq \lambda \leq 1$.

4. In the gradient descent algorithm, $\mathbf{a}_{k+1}$ is obtained from $\mathbf{a}_k$ by

$$\mathbf{a}_{k+1} = \mathbf{a}_k - \rho_k \boldsymbol{\nabla} J(\mathbf{a}_k),$$

   where $\rho_k$ is a positive scale factor that sets the step size. Consider the criterion function

$$J_q(\mathbf{a}) = \sum_{\mathbf{y} \in \mathcal{Y}} (\mathbf{a}^T \mathbf{y} - b)^2$$

   where $\mathcal{Y}(\mathbf{a})$ is the set of samples for which $\mathbf{a}^T \mathbf{y} \leq b$. Suppose that $\mathbf{y}_1$ is the only sample in $\mathcal{Y}(\mathbf{a}_k)$. Show that $\boldsymbol{\nabla} J_q(\mathbf{a}_k) = 2(\mathbf{a}_k^T \mathbf{y}_1 - b)\mathbf{y}_1$ and that the matrix of second partial derivatives is given by $D = 2\mathbf{y}_1 \mathbf{y}_1^T$. Use this to show that when the optimal $\rho_k$ is used in the gradient descent algorithm,

$$\mathbf{a}_{k+1} = \mathbf{a}_k + \frac{b - \mathbf{a}^T \mathbf{y}_1}{||\mathbf{y}_1||^2}\mathbf{y}_1.$$

5. Show that the partial derivatives of the functions $y_i = \exp(a_i)/\sum_k \exp(a_k)$ used in multiple class logistic discrimination are given by

$$\frac{\partial y_i}{\partial a_j} = y_i(\delta_{ij} - y_j)$$

   where $\delta_{ij} = \begin{cases} 1, & i = j, \\ 0, & \text{otherwise.} \end{cases}$

## Data Assignment (3 points)

6. In Module 1 we introduced nine (9) datasets. The data files are posted here in Module 9. Choose three (3) of these datasets and for each of the three:

   (a) i. Plot the data using a different color for each class.
   ii. Using the known distributions, plot the Bayes decision boundaries.

   (b) Split each dataset into 50% training data and 50% testing data.
   i. Plot the testing data using a different color for each class.
   ii. Compute estimates for the parameters (mean and covariance) of each class.
   iii. Plot the decision boundary.
   iv. Use this classifier to classify the testing data.
   A. Plot the decision boundary and the testing data.
   B. Label misclassified points.
   C. How well did the classifer do? Compute the percentage correct.